

(a) Rollout & graph

Policy rollout

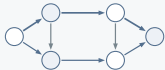
$s_0 \rightarrow s_1 \rightarrow \dots \rightarrow s_T$

rewards    

K costs    

↓ build graph

Empirical graph



$\hat{F}, A, C_k, \mu_0$

$\bar{J}_k$ (episode costs)

GAE advantages

(b) Unified price construction

Joint graph solve



$$b_k = B_k/L$$
$$F_H = 0$$

$$\min_{F \geq 0} -\langle A, F \rangle + \varepsilon KL(F \| \hat{F})$$

$$F1 - \gamma F^T 1 = (1-\gamma)\mu_0$$

$$\langle C_k, F \rangle \leq b_k \text{ (soft channels)}$$

Newton in u

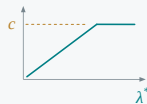
Bisection in λ

Episode-cost feedback

$$\beta_k \leftarrow [\beta_k + \eta(\bar{J}_k - B_k)]_+$$

Clip + filter

$$\lambda^- \leftarrow (1-\alpha)\lambda^- + \alpha \min(\lambda^*, c)$$



β

+

$$\lambda_k = \lambda_k^- + \beta_k$$

one price vector

(c) Policy update

Priced advantage

$$\tilde{A} = \frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k}$$

PPO / TRPO
policy step

π_{t+1}

GAE: A_r, A_{c_k}

next rollout